Programming Using Numbers (part 2)

CMSC 104 Section 2
February 11, 2026

Administrative Notes

- Should have the sample quiz out**
- Quiz 1 is next Wednesday - Feb 18, in class
- Project 1 is out today
    - We’ll talk about it at the end of class
    - It includes material we’ll cover today
    - You have two weeks to do it - the due date is Feb 25 before midnight
- Classwork 3 will be assigned next Monday, and due February 23 as well

Importing and using the Math library

The first part of this lecture involves loading and using Python libraries

- You’ll need that for Project 1

Then we’ll cover Boolean variables, which you need in Classwork 3

Python libraries

There are literally thousands of Python libraries available for free use

- This is code that others have written, and made available for free
- See e.g. https://pypi.org
- Some libraries come standard with your Python download; you just have to import them into your program.
  - Others have to be retrieved from sites like pypi.org before you can import them
- One of these is the math library. See https://docs.python.org/3/library/math.html for details

Importing libraries

The easiest way to import a library is:

```
import math
```

This line will *usually* be at or near the type of your Python code. If you try to use code from a library before you import the library, your program will crash. So you put the import line near the top, to prevent that type of mistake.

“math” is a pretty short name. But some libraries have longer names, and you might want to use a shorter nickname for it. You do that like this:

```
import math as m
```

Now instead of using “math.sqrt” you can just say “m.sqrt.”

Using functions from the math library

After you've imported the library, you can use any function in it. You'll use square root (math.sqrt()) in Classwork 3. You'll use cube root (math.cbrt()) in Project 1.

```
import math
num = int(input("Please enter a positive integer: "))
print(“The square root of :”, num, “ is “, math.sqrt(num))
```

```
next_num = int(input("Please enter any integer:
"))
print("The cube root of ", next_num , " is equal
to ", math.cbrt(next_num))
```

Other functions

Print the value of pi:

```
print(math.pi)
```

Calculate e to a given power:

```
n = int(input("Please enter a number: "))

print(math.exp(n))
```

Nicknames

I said you can use a shorter name if you want:

```
import math as m
n = float(input("Please enter a number: "))
print(m.exp(n))
```

Boolean variables

Boolean variables are Python variables that have one of two possible values: True or False

True and False are reserved words in Python

- The case is important - the first letter must be uppercase and the rest of the word must be lowercase.
- Any other combination is not the Boolean value; it’s a string. And that’s guaranteed to mess up your program.

Why are they important?

They let us control conditional action in the program.

We can do things if the value of a Boolean variable is True; and not if it's False.

Examples of Boolean variables

```
x = True

y = False

z = not x

alpha = (5 < 3+3)

beta = (4 % 3 > 4//3)

gamma = (4%3 < 4/3)
```

The “if-else” statement

The most common usage of Boolean variables is in a statement called “if - else.”

This is used when you want to do something only if a condition is True; otherwise, you don’t do it.

“If you are 18 or older, you can vote in an election. Otherwise you’re too young.”

“If your grade is “A” or “B” in a class, you can take the next class in the sequence. Otherwise you have to repeat a class.”

Some if - else examples

```
if y:
   print("Yes")
else:
   print("No")


if (40%3 == 0):
   print("Yes")
else:
   print("No")
```